

Giancarlo Giannetti

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Experience

NetProfitXL, Toronto — Software Developer

May 2022 – August 2023

- Developed back-end features in Python and SQL for ERP systems, customizing modules for client workflows.
- Built front-end interfaces with HTML, JavaScript, Jinja, and Zebra for documents, fillable forms, and webpages.
- Worked directly with clients on-site to determine system requirements.

Education

Ontario Tech University, Oshawa

Master's Student, Computer Science (MSc)

May 2024 – Present

- Expected completion: February 2026
- Research focus on Hyperspectral Unmixing using transformer-based variational autoencoders.

Ontario Tech University, Oshawa

Honours Bachelor of Science, Computer Science (BSc)

September 2019 – December 2023

- Minor in Mathematics

Projects

LDVAE-T — Project Link

Developed a Transformer-based Variational Autoencoder with a Dirichlet-constrained latent space for hyperspectral image unmixing, enforcing physical abundance constraints for more accurate material separation. Achieved state-of-the-art performance on benchmark datasets.

VCLab-ShazNet — Project Link

Built a multi-class Transformer-based audio classification system for drone detection and identification. Designed for the 2025 Defense-Tech Hackathon and awarded 2nd place overall with best technical implementation.

Skills & Awards

Skills

- Python, Java, C#, C++, Kotlin, SQL, HTML, JavaScript, VBA, C, Flutter, Jinja
- PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, CUDA, Pandas
- Git, Docker, Virtual Machines

Awards

- 1st Place - UOttHack8 SurveyMonkey sponsor challenge - Project Link
- 2nd Place - Defense-Tech Hackathon 2025
- 3× Dean's List, 2× President's List